

```
from flask import Flask, render_templatestring, request, jsonify
```

```
app = Flask(name)
```

```
users = {}
```

```
plans = {}
```

```
progress = {}
```

Frontend HTML template

```
HTML_TEMPLATE = """
```

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8"/>
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0"/>
```

```
<title>AI Health & Fitness Planner</title>
```

```
<style>
```

```
* { box-sizing: border-box; margin: 0; padding: 0; }
```

```
body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; background: #f4f6f8; color: #333; }
```

```
header { background: #4CAF50; color: white; text-align: center; padding: 20px; font-size: 1.8rem; font-weight: bold; }
```

```
nav { display: flex; background: #e0e0e0; border-bottom: 1px solid #ccc; flex-wrap: wrap; }
```

```
nav button { flex: 1; border: none; background: transparent; padding: 15px 10px; cursor: pointer; font-size: 1rem; transition: 0.3s; }
```

```
nav button.active, nav button:hover { background: #4CAF50; color: #fff; font-weight: bold; }
```

```
.container { max-width: 700px; margin: 25px auto; background: white; padding: 25px; border-radius: 12px; box-shadow: 0 4px 10px rgba(0,0,0,0.1); }
```

```
input, select, button, textarea { width: 100%; margin-top: 12px; padding: 12px; font-size: 1rem; border-radius: 6px; border: 1px solid #ccc; }
```

```
button.primary { background: #4CAF50; color: white; border: none; border-radius: 6px; cursor: pointer; transition: 0.3s; }
```

```
button.primary:hover { background: #45a049; }
```

```
textarea { height: 150px; resize: none; background: #f9f9f9; }
```

```
.hidden { display: none; }
```

```
.plan-card { background: #fdfdfd; margin-bottom: 15px; padding: 15px; border-radius: 8px; border-left: 6px solid #4CAF50; box-shadow: 0 2px 5px rgba(0,0,0,0.05); }
```

```
.plan-card h4 { margin-bottom: 8px; color: #333; font-size: 1.1rem; }
```

```
.plan-card p { font-size: 0.95rem; color: #555; }
```

```
.history-item { border-bottom: 1px solid #eee; padding: 10px 0; font-size: 0.95rem; }
```

```
.progress-list label { display: block; margin-bottom: 8px; font-size: 0.95rem; }
@media (max-width: 500px) { nav button { font-size: 0.9rem; padding: 12px 5px; } }
</style>
</head>
<body>
```

```
<header>AI Health & Fitness Planner</header>
```

```
<nav>
<button class="active" onclick="switchTab('loginTab',this)">Login</button>
<button onclick="switchTab('planTab',this)">Planner</button>
<button onclick="switchTab('historyTab',this)">History</button>
<button onclick="switchTab('progressTab',this)">Progress</button>
</nav>
```

```
<!-- LOGIN / SIGNUP -->
<div id="loginTab" class="container">
<h3>Login / Signup</h3>
<input type="text" id="name" placeholder="Name (for signup)">
<input type="email" id="email" placeholder="Email">
<input type="password" id="password" placeholder="Password">
<button class="primary" onclick="signup()">Signup</button>
<button class="primary" onclick="login()">Login</button>
<p id="authStatus" style="margin-top:12px;color:#4CAF50;font-weight:bold;"></p>
</div>
```

```
<!-- PLAN GENERATION -->
<div id="planTab" class="container hidden">
<h3>Generate Personalized Plan</h3>
<input type="text" id="goal" placeholder="Goal (e.g., Weight Loss)">
<select id="fitness_level">
<option>Beginner</option>
<option>Intermediate</option>
<option>Advanced</option>
</select>
<input type="number" id="workout_days" placeholder="Workout Days (e.g., 3)">
<button class="primary" onclick="generatePlan()">Generate Plan</button>
<div id="planDisplay" style="margin-top:15px;"></div>
</div>
```

```
<!-- HISTORY -->
<div id="historyTab" class="container hidden">
```

```
<h3>Your Past Plans</h3>
<div id="historyList"></div>
</div>
```

```
<!-- PROGRESS -->
<div id="progressTab" class="container hidden">
<h3>Workout Progress</h3>
<div class="progress-list" id="progressList"></div>
<button class="primary" onclick="saveProgress()">Save Progress</button>
</div>
```

```
<script>
let userId = null;
let currentPlan = null;

function switchTab(tabId, btn) {
document.querySelectorAll('nav button').forEach(b => b.classList.remove('active'));
btn.classList.add('active');
document.querySelectorAll('.container').forEach(c => c.classList.add('hidden'));
document.getElementById(tabId).classList.remove('hidden');
if(tabId === 'historyTab') loadHistory();
if(tabId === 'progressTab') loadProgress();
}


```

```
async function signup(){
const res = await fetch('/signup', { method:'POST', headers:{'Content-Type':'application/json'},
body: JSON.stringify({
name: document.getElementById('name').value,
email: document.getElementById('email').value,
password: document.getElementById('password').value
}}));
const data = await res.json();
document.getElementById('authStatus').innerText= data.message;
}


```

```
async function login(){
const res = await fetch('/login', { method:'POST', headers:{'Content-Type':'application/json'},
body: JSON.stringify({
email: document.getElementById('email').value,
password: document.getElementById('password').value
}}));
const data = await res.json();


```

```

if(data.status === 'success'){
  userId = data.user_id;
  document.getElementById('authStatus').innerText= "Logged in successfully!";
  switchTab('planTab',document.querySelectorAll('nav button')[1]);
} else {
  document.getElementById('authStatus').innerText= data.message;
}
}

```

```

async function generatePlan(){
  if(!userId) { alert("Login first!"); return; }
  const display = document.getElementById('planDisplay');
  display.innerHTML = "Generating plan...";
  const res = await fetch('/generate-plan',{ method:'POST',
  headers: {'Content-Type':'application/json'},body: JSON.stringify({
  user_id: userId,
  goal: document.getElementById('goal').value,
  fitnesslevel: document.getElementById('fitnesslevel').value,
  workoutdays: parseInt(document.getElementById('workoutdays').value)
  })));
  const data = await res.json();
  displayPlan(data.plan);
  currentPlan = data.plan;
}

```

```

function displayPlan(planObj){
  const display = document.getElementById('planDisplay');
  display.innerHTML = "";
  if(planObj.workout_plan){
    for(let day in planObj.workout_plan){
      const card = document.createElement('div');
      card.className='plan-card';
      card.innerHTML = `

#### ${day}</h4><p>${planObj.workout_plan[day]}</p>`; display.appendChild(card); } } if(planObj.mealplan){ display.innerHTML += ` Generated with https://kome.ai


```

```

async function loadHistory(){
  if(!userId) return;
  const res = await fetch(`/history/${userId}`);
  const data = await res.json();
  const list = document.getElementById('historyList');
  list.innerHTML = data.history.map(h=> `<div class="history-item"><b>${h.created
_at}</b><pre>${h.plan}</pre></div>`).join("");
}

```

```

async function loadProgress(){
  if(!userId) return;
  const res = await fetch(`/progress/${userId}`);
  const data = await res.json();
  const container = document.getElementById('progressList');
  container.innerHTML="";
  (currentPlan?.workoutplan ? Object.keys(currentPlan.workoutplan) :
  ['Day1','Day2','Day3']).forEach(day => {
    container.innerHTML += `<label><input type="checkbox" value="${day}"
    ${data.progress.find(p=>p.day===day && p.status==='completed')?'checked':''}>/>
    ${day}</label>`;
  });
}

```

```

async function saveProgress(){
  if(!userId) return;
  const checkboxes = document.querySelectorAll('#progressList input[type="checkbox"]');
  for(const cb of checkboxes){
    await fetch('/progress', {
      method:'POST',
      headers:{'Content-Type':'application/json'},
      body: JSON.stringify({user_id:userId, day:cb.value, status: cb.checked ?
      "completed":"pending"})
    });
  }
  alert('Progress saved!');
}
</script>

```

</body>

</html>

.....

```
@app.route('/')
def home():
    return render_templatestring(HTML_TEMPLATE)
```

```
@app.route('/signup', methods=['POST'])
def signup():
    data = request.get_json()
    email = data['email']
    if email in users:
        return jsonify({'message': 'User already exists', 'status':'fail'})
    users[email] = {'name': data['name'], 'password': data['password']}
    return jsonify({'message': 'Signup successful!', 'status':'success'})
```

```
@app.route('/login', methods=['POST'])
def login():
    data = request.get_json()
    email = data['email']
    password = data['password']
    if email in users and users[email]['password'] == password:
        return jsonify({'status':'success','user_id': email})
    return jsonify({'status':'fail','message':'Invalid credentials'})
```

```
@app.route('/generate-plan', methods=['POST'])
def generate_plan():
    data = request.get_json()
    userid = data['userid']
    plan = {
        "workoutplan": { f"Day {i+1}": f"{data['goal']} - {data['fitnesslevel']} workout" for i in
            range(data['workout_days']) },
        "meal_plan": "Balanced meals with protein, carbs, and veggies.",
        "motivation": "Stay consistent and never give up!"
    }
    plans.setdefault(user_id, []).append(plan)
    return jsonify({'plan': plan})
```

```
@app.route('/history/<user_id>')
def history(user_id):
    userhistory = plans.get(userid, [])
    return jsonify({'history': [{'createdat': 'Today', 'plan': str(p)} for p in userhistory]})
```

```
@app.route('/progress/<user_id>') 'is this for frontend prototype code for my app
```